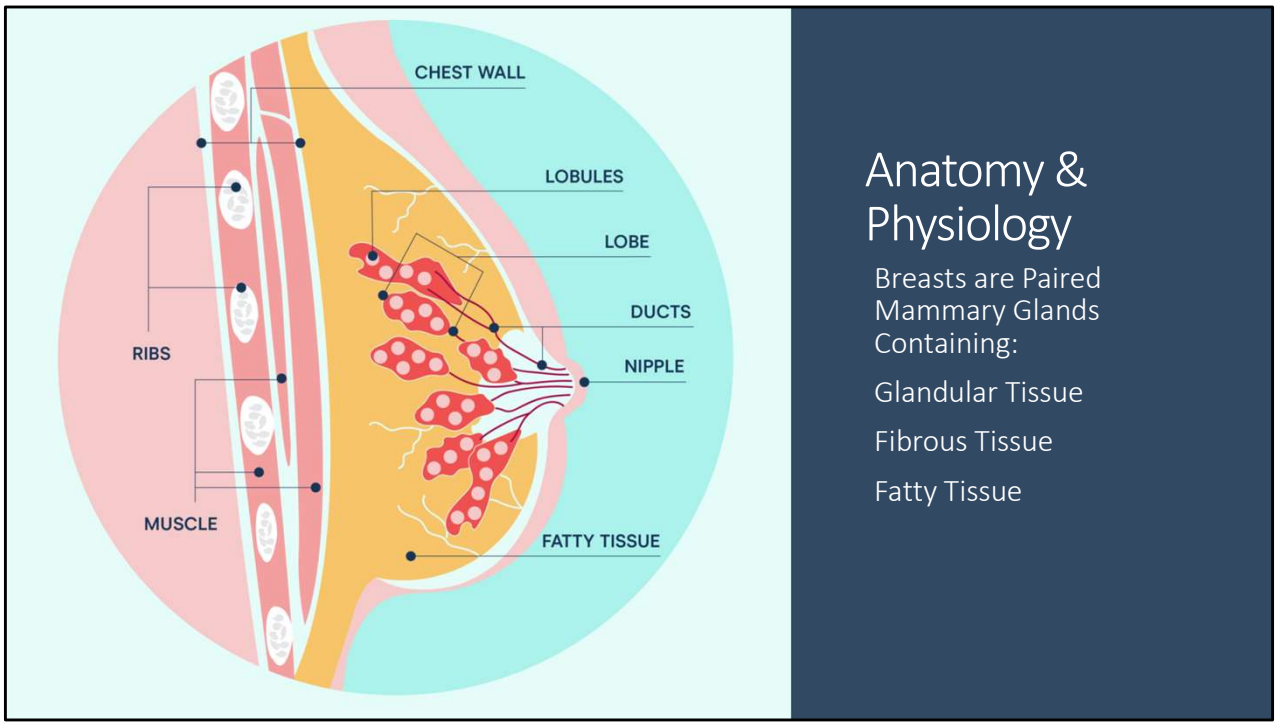
# Assessing breasts & lymphatic system

Weber: Chapter 20

NURS3120: Health Assessment and Promotion

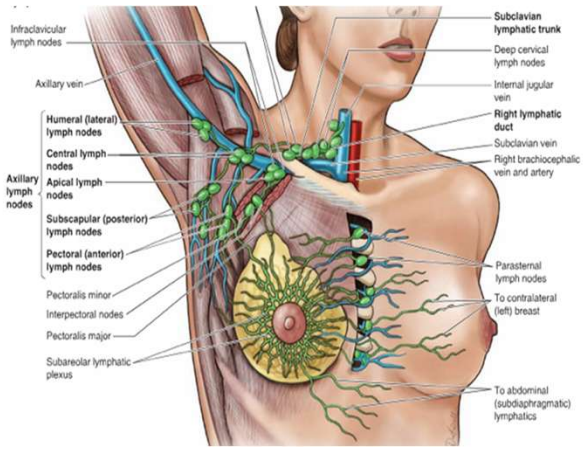
# Objectives

1. Describe the structure and function of the breast and major axillary lymph nodes.
2. Discuss risk factors associates with breast cancer and ways to reduce one's risks.
3. Interview a client for an accurate nursing history of the breasts and axillary lymph nodes.
4. Differentiate between normal and abnormal findings of breasts and axillary lymph nodes.




As you remember from anatomy and physiology these are the main parts of the breast.
Female breasts consist of three types of tissue: glandular, fibrous, and fatty (adipose). Glandular tissue constitutes the functional part of the breast, allowing for milk production. Glandular tissue is arranged in 15 to 20 lobes that radiate in a circular fashion from the nipple. Each lobe contains several lobules in which the secreting alveoli (acini cells) are embedded in grape-like clusters.
Mammary ducts from the alveoli converge into a single lactiferous duct that leaves each lobe and conveys milk to the nipple.
Fibrous tissue: provides support to the breast.
Fatty tissue: The glandular tissue is embedded in the fatty tissue. This subcutaneous and retromammary fat provides most of the substance to the breast, determining the size and shape of the breasts. The functional capability of the breast is not related to size but rather to the glandular tissue present.

# Lymphatics




Breast have extensive drainage into the axillary nodes.

- Central (midaxillary)
- Anterior (Pectoral)
- Posterior (Subscapular)
- Lateral (Brachial)

Lymph nodes are present in both male and female breasts. These structures drain lymph from the breasts to filter out microorganisms and return water and protein to the blood.

The anterior nodes drain the anterior chest wall and breasts. The posterior chest wall and part of the arms are drained by the posterior nodes.

The lateral nodes drain most of the arms, and the central nodes receive drainage from the anterior, posterior, and lateral lymph nodes.

## Collecting Subjective Data

- Any breast Pain?
- Any lumps, swelling, or thickening?
- Any discharge? (Galactorrhea)
- Any “breast self-awareness” techniques?

With breast pain always use COLDSPA. Could be fibrocystic breasts, inflammation, malignancy…
Lumps, swelling, thickening could be fibrocystic breast disease, fibroadenomas, malignancy, infection…
A client reporting discharge that is spontaneous, unilateral and blood or blood-tinged discharge needs to be further evaluated. Sometimes clear benign discharge may be manually expressed from a breast that is frequently stimulated. Certain medications are also associated with clear discharge like oral contraceptives, phenothiazines, steroids, digitalis, and diuretics. Galactorrhea is usually medication induced or due to a pituitary tumor.

The American College of Obstetrics and Gynecology (ACOG) no longer recommends breast self-exam for individuals at average risk for breast cancer. Many organizations now advise an alternative called “breast self-awareness” (becoming familiar with the appearance, feel, and shape of one’s breasts and nipples).

Breast self-exam is an option for those at high risk for breast cancer. Individuals should report any breast changes to their provider.

# Assessment

- Inspection
- Skin color & texture
- Size & symmetry
- Discharge from nipples
- Retraction & dimpling
- Nipple inversion or retraction
- Peau D'Orange

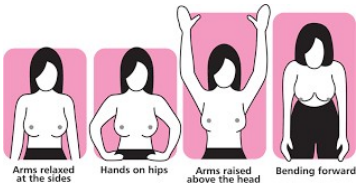



Breasts can be a variety of sizes and are somewhat round and pendulous. One breast might be larger than the other. These are normal findings, but a recent increase in the size of one breast may indicate inflammation or an abnormal growth.
Inspect Color and texture, areolas, and nipples are nearly equal bilaterally in size and are in the same location.
Color varies depending on the client's skin tone. Texture is smooth, with no edema. Linear stretch marks may be seen during and after pregnancy or with significant weight gain or loss.
The client's breasts should rise symmetrically, with no sign of dimpling or retraction. Breasts should hang freely and symmetrically.

A recent retracted nipple that was previously everted suggests malignancy.
Inspect for dimpling or retraction: This is an abnormal finding and is usually caused by a malignant tumor that has fibrous strands attached to the breast tissue and the fascia of the muscles. As the muscle contracts, it draws the breast tissue and skin with it , causing dimpling or retraction.
Peau D'Orange is when the skin turns the color orange, and the texture also takes on

that of an orange peel. Results from edema caused by blocked lymphatic drainage. Seen in inflammatory breast cancer.

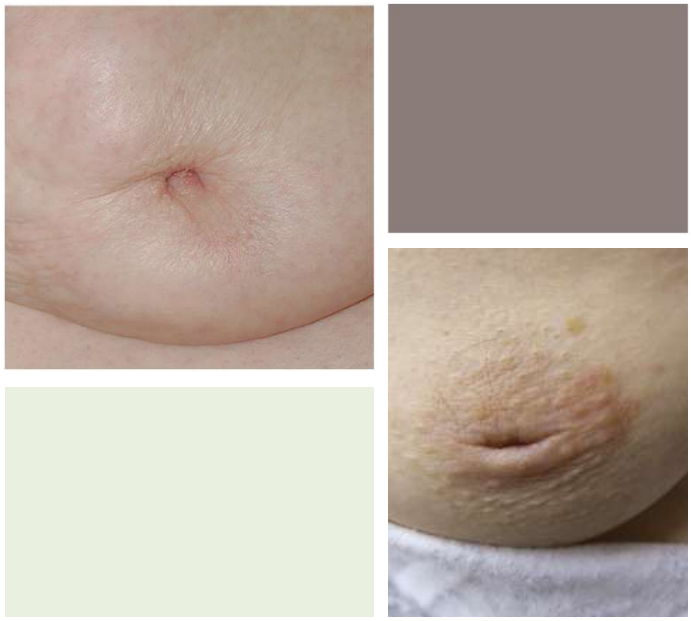
# Inspect for Retraction & Dimpling




1. Hands over head
2. Hands on hips
3. Push palms together
4. Lean forward with forearms supported for large breasts

This allows for observation of any presence of retraction

During assessment for retraction and dimpling, the client first raises her arms over her head, then lowers them and presses them against the hips and finally presses the hands together with fingers of one hand pointing opposite to the fingers of the other hand and finally ask the client to lean forward from the waist. You are inspecting for restricted movement of the breast or retraction of the skin or nipple which could indicate fibrosis and fixation of the underlying tissues. This could be due to a tumor.



## Nipple inversion vs. retraction

Inversion is usually congenital and not concerning

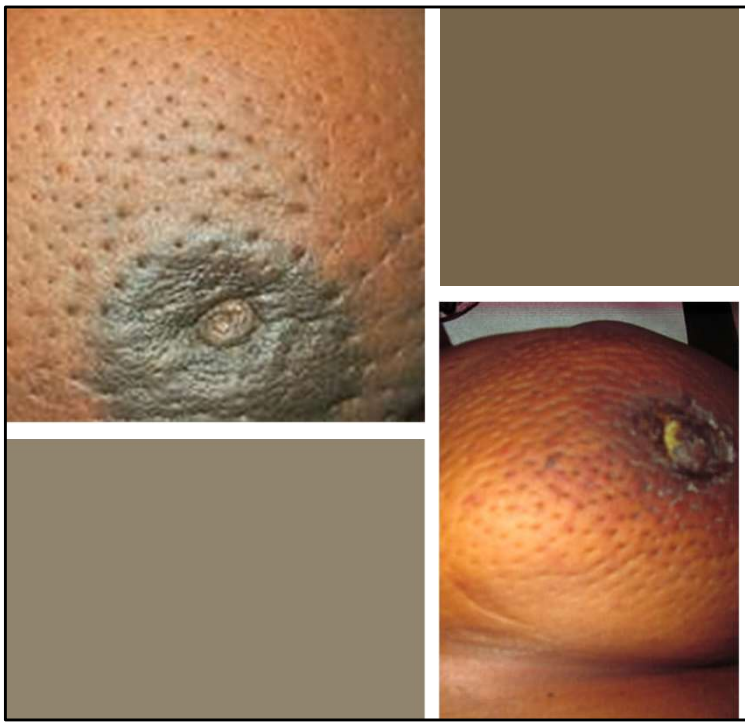
Any retraction or inversion that is **NEW** needs to be evaluated

(Malignant retraction on left, congenital inversion on right)

Nipple inversion occurs as a normal finding in approximately 3% of women. It is often congenital, although it can be acquired in cases of trauma, scarring mastitis, breast cancer, and breast surgeries. In most congenital cases, the nipple will protract during stimulation, allowing for normal breastfeeding.

Nipple retraction or inversion that is **new**, occurring only in one breast, or where the nipple does not come out on stimulation can be a concerning sign of breast cancer.

Nipples that are inverted or retracted point inward toward the breast or lay flat instead of pointing outward. Most people with inverted nipples are born with them. And usually, they're harmless. If one or both of your nipples inverts suddenly, though, this could be a sign of an issue that requires medical attention.



## Abnormal Findings of the Breast on Inspection

**Peau D'Orange**– "Orange peel" associated with inflammatory breast cancer

**Paget Disease** – Rare breast cancer of the nipple & areola

**Retracted Nipple-** suggest malignancy

**Dimpling**- suggest malignancy

**Retracted Breast Tissue-** suggest malignancy

**Mastitis**- reddened, painful area on breast r/t infection

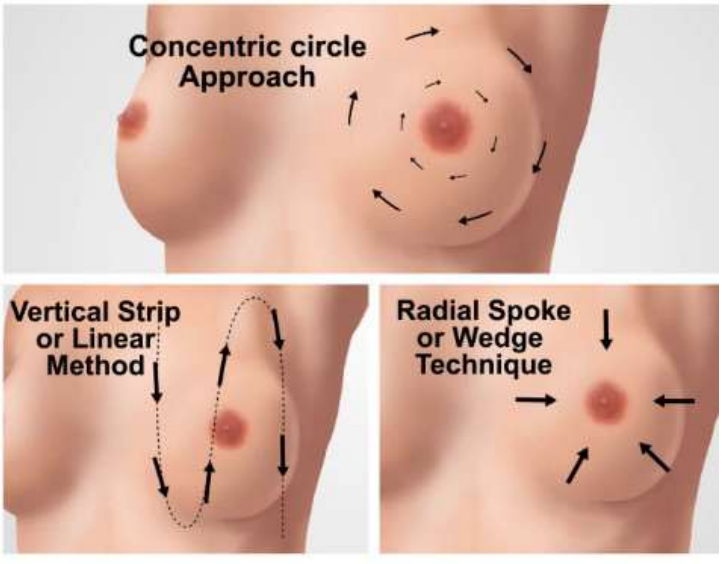
**Paget Disease** –
**Peau D'Orange: Inflammatory Breast cancer (IBC):** Cancer cells block lymph vessels, causing swelling and pitting, often with redness, warmth, and thickening of the breast skin

Paget's disease of the breast is a rare breast cancer affecting the nipple and areola, appearing as a scaly, itchy, red rash often mistaken for eczema, but signals underlying cancer (DCIS or invasive). Symptoms include crusting, flaking, nipple flattening, or discharge, usually in one breast



# Abnormal Findings on Inspection

Skin dimpling and retraction on left
Paget Disease on right




## Palpation

- 3 patterns:
  - Concentric circle
  - Vertical strip
  - Radial spoke
- Palpate for:
  - Tenderness
  - Temperature
  - Texture
  - Masses
  - Nipple for discharge

When palpating the breast ask the client to lie down and to place overhead the arm on the same side as the breast being palpated. Place a small pillow or rolled towel under the breast/back while being palpated. Use the flat pads of three fingers to palpate the client's breasts. Using one these three patterns. Be sure to palpate every square inch of the breast from the nipple and areola to the periphery of the breast tissue and up the tail of Spence. Vary the levels of pressure: light, medium, and firm. Vertical is up and down starting on one side of the breast and working across to the other side, Wedge looks like a spokes of a wheel pattern; You start from the nipple and work outward and circular is you make a circular pattern and start from the nipple and work all the way around the breast until you have completed the whole breast.

**During palpation**: palpate for texture, elasticity, tenderness, temperature: the skin should be smooth, firm and elastic tissue, breasts should be normal body temperature, and no masses or tenderness should be noted, unless the client is having a menstrual cycle or on hormonal medications, then some generalized tenderness might be noted. Abnormal: heat noted from the skin temperature might indicate a malignant tumor.

**Palpate the Nipples:** wearing gloves gently compress the nipple with your thumb and index finger



## The Axillae

Inspection: Rashes, infection, swelling, inflammation, or change in coloring

Palpation: Midaxillary, supraclavicular, anterior, and posterior nodes

Have the client sit up and slightly raise the client's arm holding it by the elbow to palpate the axillary lymph nodes. First palpate high into the axillae, moving downward against the ribs to feel for the central nodes. Continue to move down the posterior axillae to feel for the posterior nodes. Use bimanual palpation to feel for the anterior axillary nodes. Finally palpate down the inner aspect of the upper arm. Normal findings include: No palpable nodes or one to two small discrete, nontender movable nodes in the central area.

Lymph nodes that are hard and fixed to the skin may indicate an underlying malignancy.

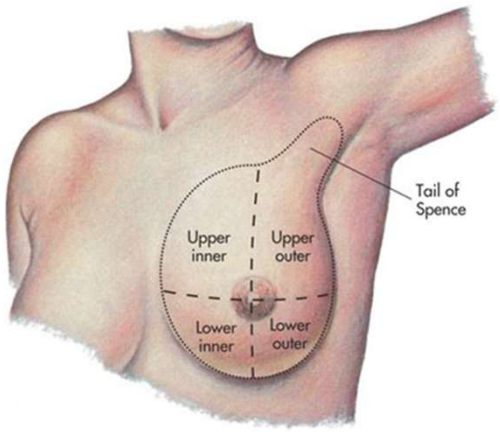


# Male Breast

- Gynecomastia
  - During adolescence breast tissue temporarily enlarges, tends to occur in approximately 70% of adolescent males
  - Reappear in the aging male
  - Changes in hormones- decreased testosterone, increase in estrogen

Gynecomastia, a smooth firm moveable disc of glandular tissue, may be seen in one breast in males during puberty, usually temporary, However, it may also be seen in hormonal imbalances, drug abuse, cirrhosis, leukemia, & thyrotoxicosis.
Irregularly shaped, hard nodules occur in breast cancer.
Just like females; men loose elasticity in their breasts as they age the to add to that medications can cause older males to develop breasts: for example, ace inhibitors, Amiodarone, Calcium channel blockers, spironolactone, diazepam, cimetidine, omeprazole, ranitidine, and hormones (decrease in testosterone)

FIGURE 16-2 Quadrants of the left breast and axillary tail of Spence.

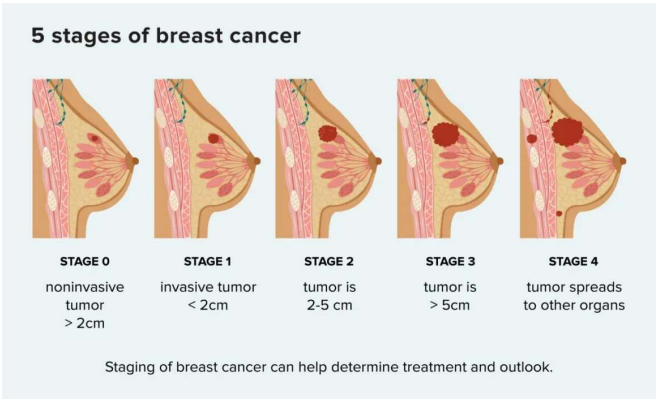



## When a Lump is Detected

**Identify the following:**

- Location
- Size
- Shape
- Consistency
- Movable?
- Distinctness (is lump solid?)
- Nipple (displaced or retracted?)
- Any redness, dimpling or retraction
- Tenderness
- Lymphadenopathy

Please contact your doctor and follow up. Always chart what you find.
The upper outer quadrant/Tail of Spence is the most common place for breast tumors.




## Evidence-based health promotion & disease prevention: Breast Cancer

**Risk factors (unmodifiable)**

- Female & >50 years
- Genetics (*BRCA1 & BRCA2*)
- Personal history of breast cancer
- Early menstruation <12 years
- >52 late menopause
- Previous chest radiation before 40yrs old

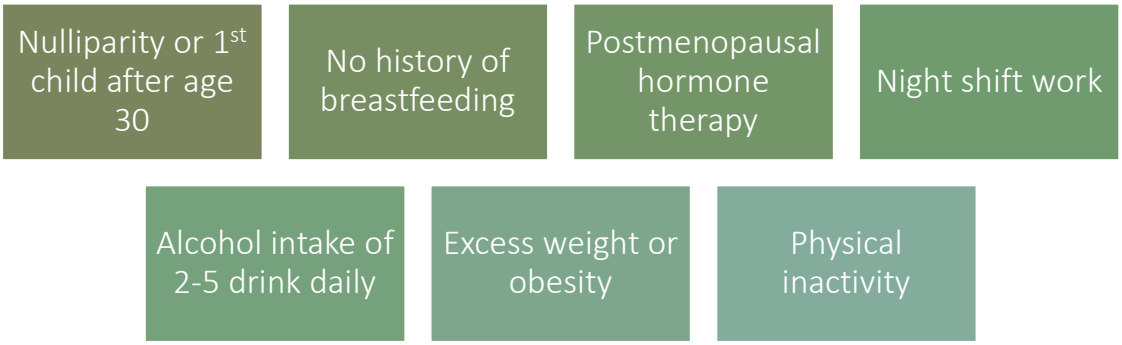
Second leading cause of cancer death (Lung cancer is first) among women in the US. It is the most common cancer diagnosed among females in the United States.

Although breast cancer is predominant mostly in females, approximately 1% of all breast cancer cases and deaths will occur among males. Breast cancer death rates in the United States have dropped by 44% since 1989 due to advances in screening, detection, and treatment.

91% 5 year survival rate per the American Cancer Society. Death rates have dropped by 44% since 1989 due to advances in screening, detection, and treatment (ACS)

Aprox 1 in 8 women will develop invasive breast cancer at some point in their life.

# Modifiable Risk Factors for Breast Cancer



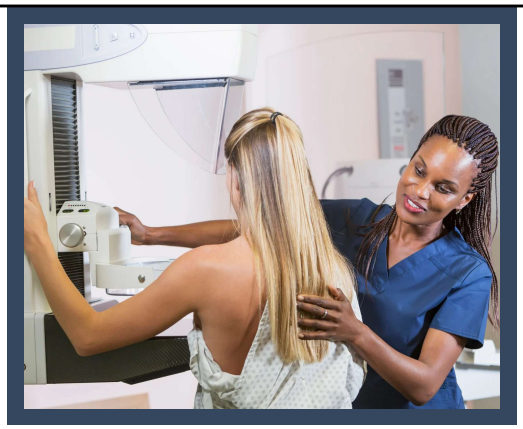

Educate clients about their risk factors!

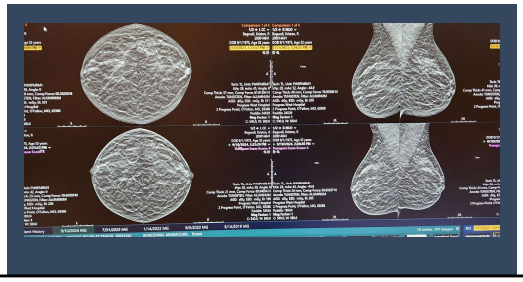
All individuals can develop breast cancer; clients should note any changes in breast size, shape, or tissue consistency and report to health care provider.
Inform clients of different screening recommendations and advise them to talk with their health care provider to determine the best screening protocol for them (see screening recommendations under Screening).

Intentional exercise: 45-60 min/day for 5 or more days a week
Avoid alcohol
Avoid excessive weight gain (especially as an adult and especially after menopause)

# Healthy People 2030 Goals

1. Reduce the female breast cancer death rate
2. Reduce late-stage female breast cancer
3. Increase the proportion of women who receive a breast cancer screening based on the most recent guidelines
4. Increase the proportion of women who were counseled by their providers about mannograms





# Breast Cancer Screening

Mammogram is screening of choice

The American Cancer Society (ACS) recommends the following:

Optional by choice between 40-44 years of age

Yearly between 45 and 54 years of age

Every 2 years age 55 and older

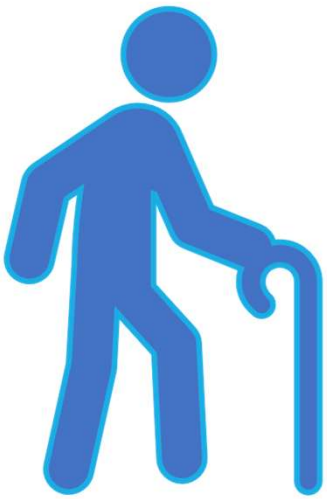
Mammogram is the screening of choice for breast cancer, and the 3D mammogram is recommended when available (Mayo Clinic, 2022b). The 3D mammogram takes multiple images to provide a three-dimensional picture of the breast.

# Fibrocystic Breast Changes

| |
|---|
| Painful, lumpy breasts |
| Worse just prior to menses |
| Improves after menses |
| Changes in size with the onset and completion of menses |
| May individuals note worsening with caffeine consumption |
| Can affect both men and women |
| Green or dark brown non-bloody nipple discharge |
| Non-cancerous |

Fibrocystic breast changes (formerly disease) are common, non-cancerous breast lumps and pain caused by hormonal fluctuations, leading to lumpy, swollen, tender breasts, especially before periods, but can often be managed with supportive bras, pain relievers, and heat/ice, though a doctor should rule out cancer and may drain painful cysts. It's not truly a disease but normal changes, affecting many women, particularly between 30 and 50, and typically .resolves after menopause.

**Hormonal fluctuations**:
Estrogen and progesterone changes during the menstrual cycle cause ducts to enlarge and retain fluid.



## Older Adult Considerations

- Atrophy of breast tissue
- Decrease in size
- Decrease in firmness
- Decrease in elasticity
- Breasts become saggy
- Smaller/flatter nipples
- Nipple inversion may be seen
- Glandular tissue decreases/Fatty tissue increases
- Axillary hair decreases

The older client may notice a decrease in the size and firmness of the breasts as she/he ages because of a decrease in estrogen levels. They often have more pendulous, less firm, and saggy breasts. The older client may have smaller flatter nipples. Nipple inversion may be seen due to fibrotic changes and shrinkage. Glandular tissue decreases whereas fatty tissue increases. A well-fitting supportive bra can reduce breast discomfort related to sagging breasts. When assessing an older client's breasts, they may feel more granular.

## Review Question

A nurse is palpating the axillary lymph nodes of a client. Which finding should the nurse recognize as abnormal?

A. Small, soft, movable nodes

B. Nodes that are non-palpable

C. Firm, fixed, non-tender nodes

D. Slight tenderness during palpation

**Correct Answer:** C
**Rationale:** Firm, fixed, non-tender lymph nodes may suggest infection or malignancy and require follow-up.

# Review Question

During a health assessment, a nurse palpates a hard, irregular, non-mobile mass in the upper outer quadrant of a client's right breast. The client denies pain. Which of the following would be next appropriate action by the nurse?

A. Reassure the client that painless lumps are usually benign

B. Ask the client to return after her next menstrual cycle

C. Document the finding and notify the health care provider

D. Teach the client proper breast self-examination technique

**Correct Answer:** C
**Rationale:** A hard, irregular, non-mobile, painless mass is a concerning finding. The nurse should document clearly and report to the provider for further evaluation.